

Surge Protection Device (SPD)

1) What is an SPD and why is it important?

A **Surge Protection Device (SPD)** is a protective device used to **protect electrical equipment from sudden voltage spikes (surges)**.

These surges can happen because of:

- Lightning strike
- Switching ON/OFF heavy motors
- DG set or transformer switching
- Utility line faults
- Sudden power restoration after outage

The surge lasts only **microseconds**, but it can damage:

- PLC
- UPS
- LED drivers
- CCTV
- Computers
- SMPS
- Industrial panels



2) Why it is important

Without SPD:

- Equipment PCB may burn
- UPS may fail
- PLC input cards can damage
- VFD trips
- Fire risk increases

SPD safely diverts the extra voltage to **earth/ground**, protecting the load.

2) How SPD works during surge or lightning strike

Normal condition

During normal voltage (230V / 415V), SPD stays **inactive (high resistance)**.

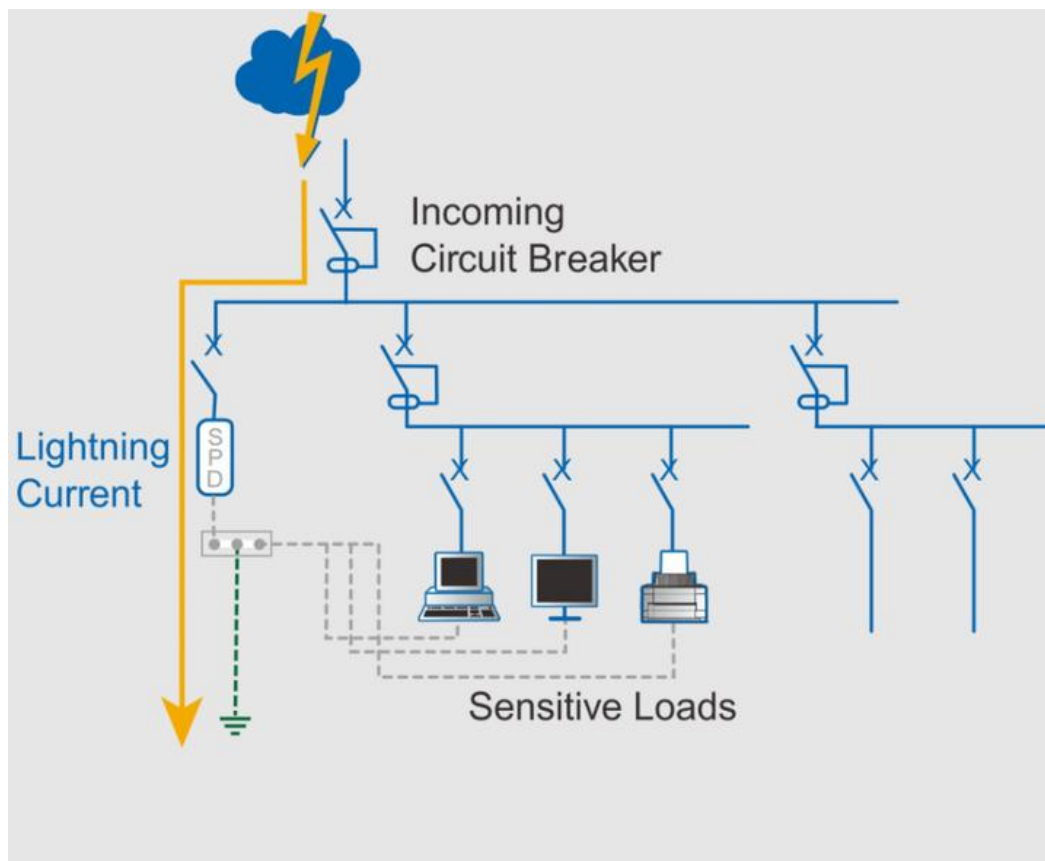
During surge

Suppose line voltage suddenly rises from **230V to 4000V** due to lightning.

The SPD instantly:

1. Detects overvoltage
2. Becomes low resistance
3. Diverts surge current to earth
4. Protects downstream equipment
5. Returns to normal automatically

Think of SPD like a **pressure relief valve for electricity**.



3) Types of SPD

◆ By internal technology

MOV (Metal Oxide Varistor)-----WE prefer this

Most common type.

- Fast response
- Good for power lines
- Used in DB and SMDB
- Best for Type 2 SPD

GDT (Gas Discharge Tube)

- Better for lightning energy
- Used in Type 1
- High surge capacity
- Used in communication/data lines too

◆ **By installation level**

Type 1 SPD

Installed at **main incoming panel / main LT panel**

Use when:

- Site has lightning risk
- Overhead line supply
- External lightning arrestor available

Best for **direct lightning surges**.

Type 2 SPD

Installed in:

- DB
- SMDB
- MCC
- UPS panel
- Floor panel

Protects against:

- Switching surges
- Indirect lightning
- Utility spikes

This is the **most commonly used SPD in buildings**.

Type 3 SPD-----WE prefer this

Installed near sensitive load:

- Computer socket
- PLC panel
- UPS output
- CCTV socket
- TV / networking devices

Provides **final level protection**.

SPD is a fast acting protective device that protects electrical and electronic equipment from lightning and switching surges by safely diverting excess voltage to earth.